CM3121 – Advanced Organic Chemistry

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Reduction reactions

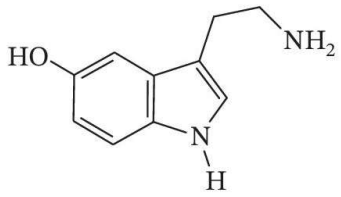
Organic molecules in everyday life

- It could even be argued that serotonin, along with dopamine were the driving forces for the evolution of mankind and even for all out scientific, artistic and cultural progress

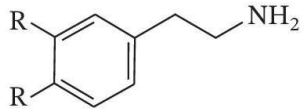
- It is found throughout the brain and central nervous system, where it helps to regulate appetite, sleep cycles, and even mood. It helps to maintain a 'happy feeling'. It also helps with memory and learning.

- Serotonin might be partly responsible for our competitive nature and constant urges to outdo our neighbors.

- Its major function is as a neurotransmitter involved in the transmission of electrical signals from one nerve to another across the synaptic gap.



Serotonin

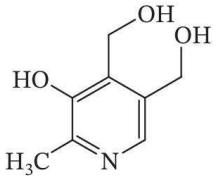


R = OH, Dopamine

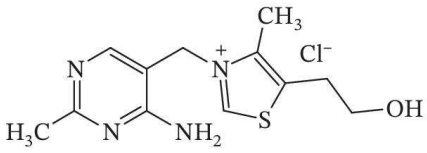
May, Paul. Molecules That Amaze Us (p. 452). CRC Press. Kindle Edition.

Nitrogen containing compounds

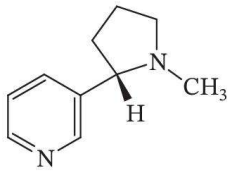
- Nitrogen containing compounds have an incredible range of biochemical properties



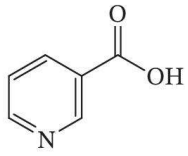
Pyridoxine (vitamin B_6)



Thiamine chloride (vitamin B_1)

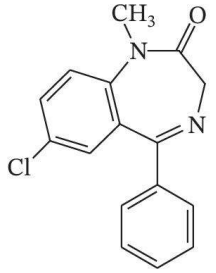


Nicotine



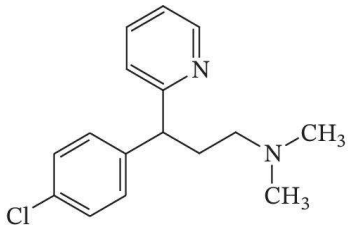
Nicotinic acid (niacin)

vitamin B3, which occurs naturally in food



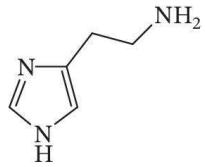
Valium (diazepam)

Diazepam is used to treat anxiety, alcohol withdrawal, and seizures



Chlorpheniramine

Anti-histamine. It is used to treat a runny nose from allergies or a cold. It is also used to treat the symptoms of an allergic reaction.



Histamine

Trimethylamine occurs in animal tissues and is partially responsible for the distinctive odor of fish

Amines

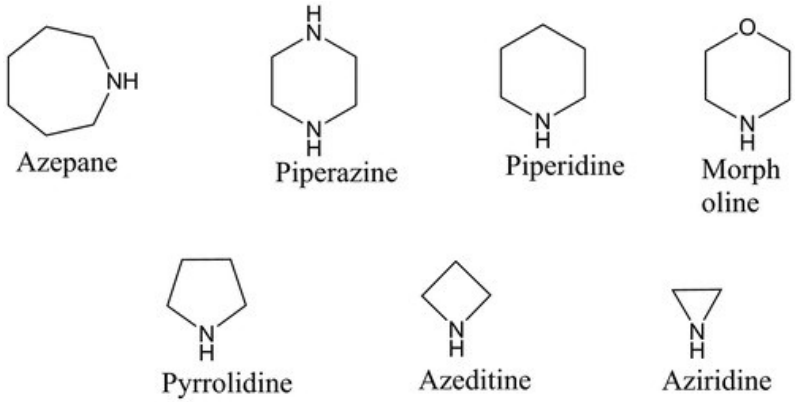
- Amines are classified as **primary (RNH_2)**, **secondary (R_2NH)**, or **tertiary (R_3N)**. Compounds containing a nitrogen atom with four attached groups also exist, but the nitrogen atom must carry a formal positive charge. Such compounds are called **quaternary ammonium salts**.

- Amines can be either alkyl-substituted (**alkylamines**) or aryl-substituted (**arylamines**). Although much of the chemistry of the two classes is similar, there are also substantial differences.

- Chemistry: Amines contain a nitrogen atom with a lone pair of electrons, making amines both basic and nucleophilic.

- In contrast with amines, amides ($RCONH_2$) are not basic. Amides aren't protonated by aqueous acids, and they are poor nucleophiles.

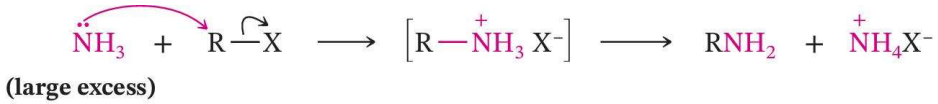
Cyclic amines



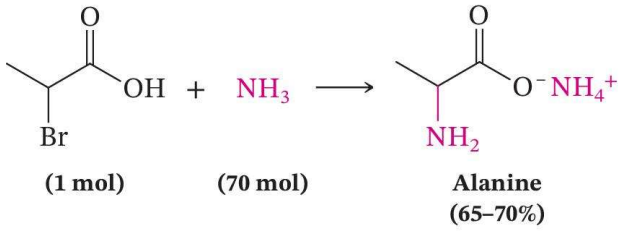
1.1 Preparation of amines: S_N2 reaction

Nucleophilic substitution reactions:

Primary amine can be prepared by ammonia and alkyl halides.

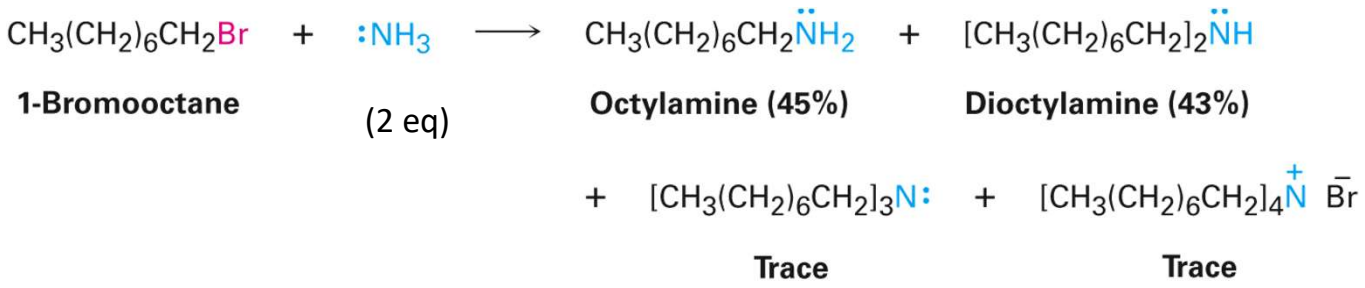


Example:



Preparation of amines

Problem:



Preparation of amines

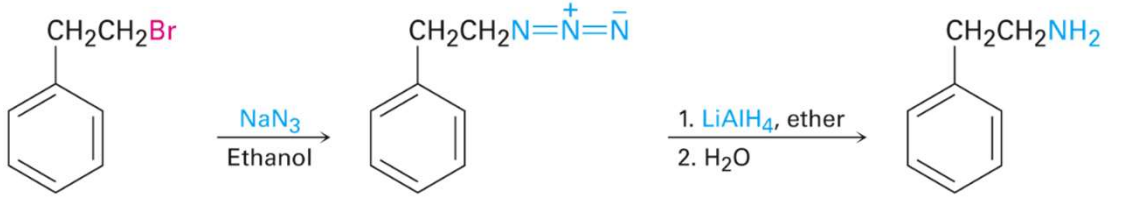
a. Alkylation of azide ion and reduction



Azide ion (a good nucleophile)

Alkyl azide

or H_2/Pd
or PPh_3



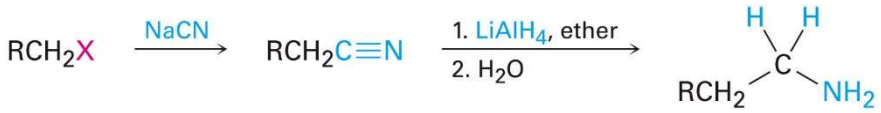
1-Bromo-2-phenylethane

2-Phenylethyl azide

2-Phenylethylamine (89%)

Preparation of amines

b. Alkylation of cyanide ion and reduction



Alkyl halide

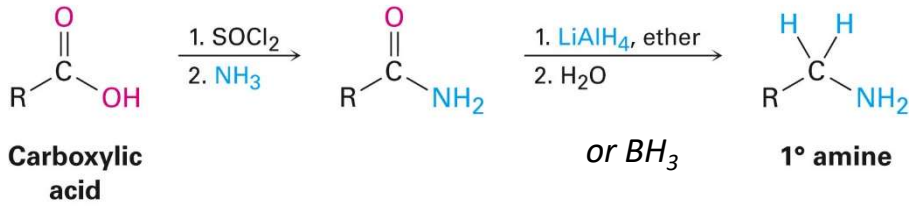
or H_2/Pd

1° amine

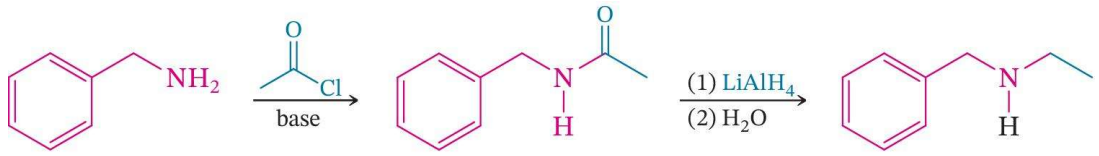
1.2 Preparation of amines: from amides

Reduction of amides

- Synthesis of primary, secondary or tertiary amines
- Reducing agents: **$LiAlH_4$ or BH_3**

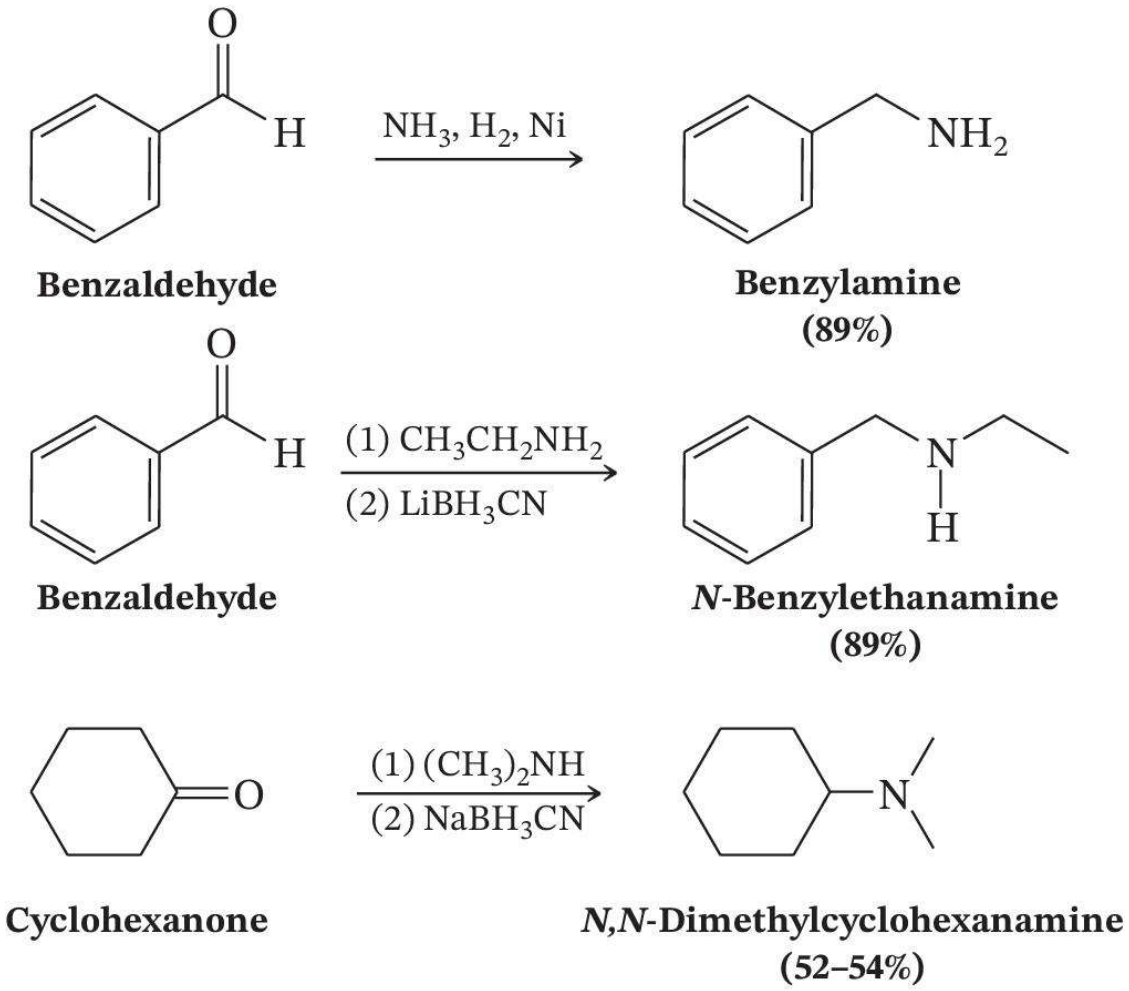


Carboxylic acid

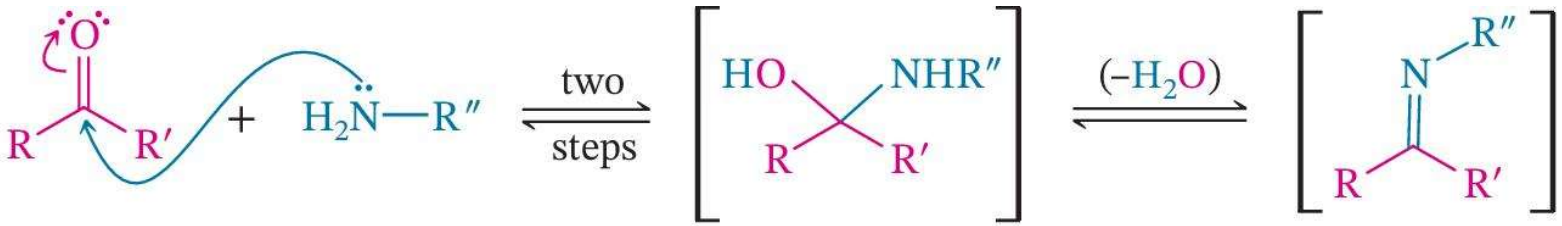


Benzylamine **Benzylethylamine**

1.3 Preparation of amine: Reductive amination



1.3 Preparation of amine: Reductive amination

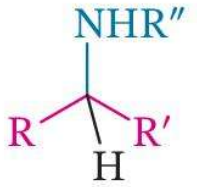


Aldehyde or ketone

1° Amine (Reaction with ammonia would lead to a primary amine. Reaction with a secondary amine would lead to a tertiary amine.)

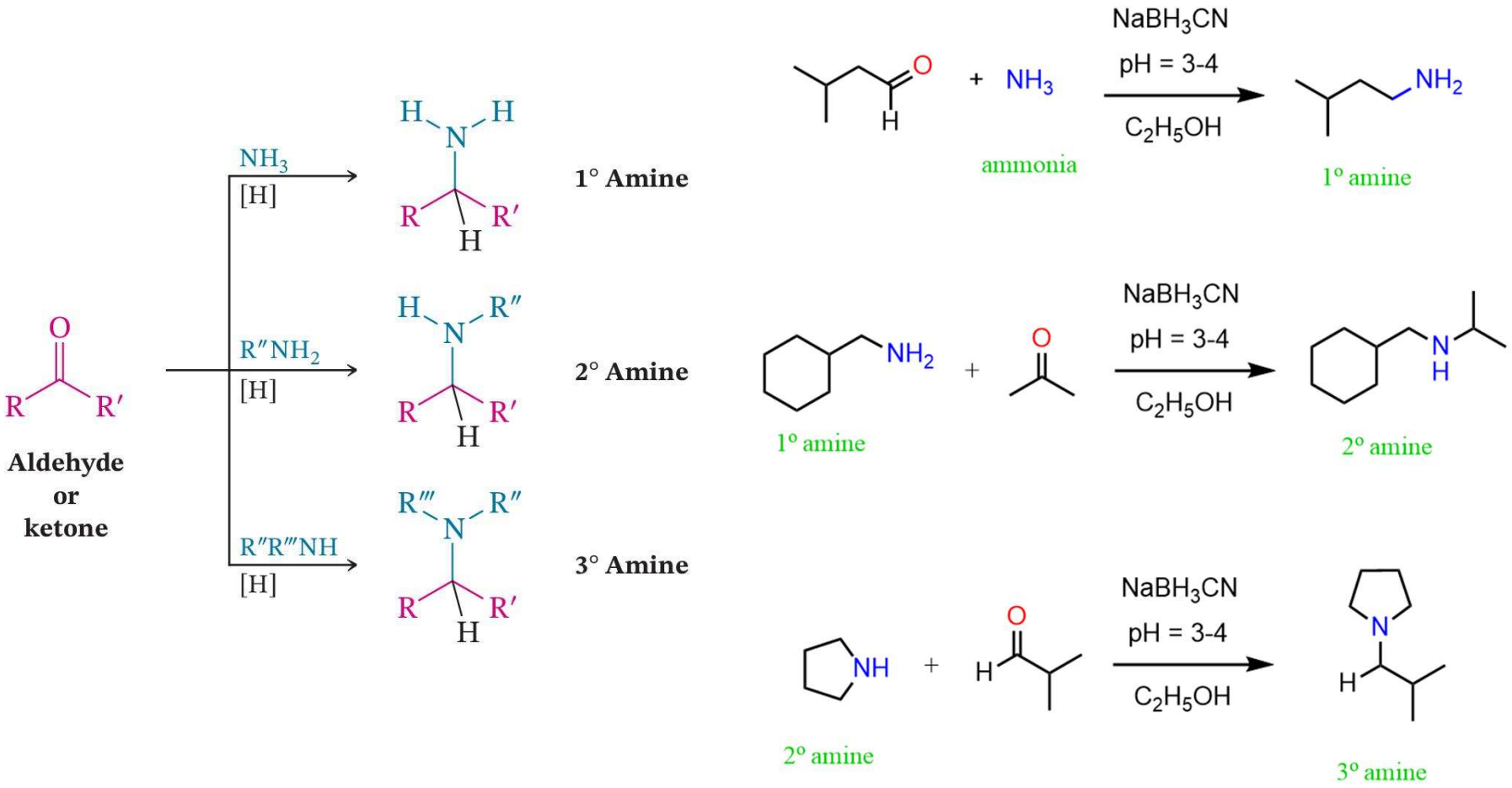
Hemiaminal

Imine



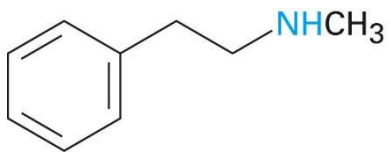
2° Amine

Preparation of amine

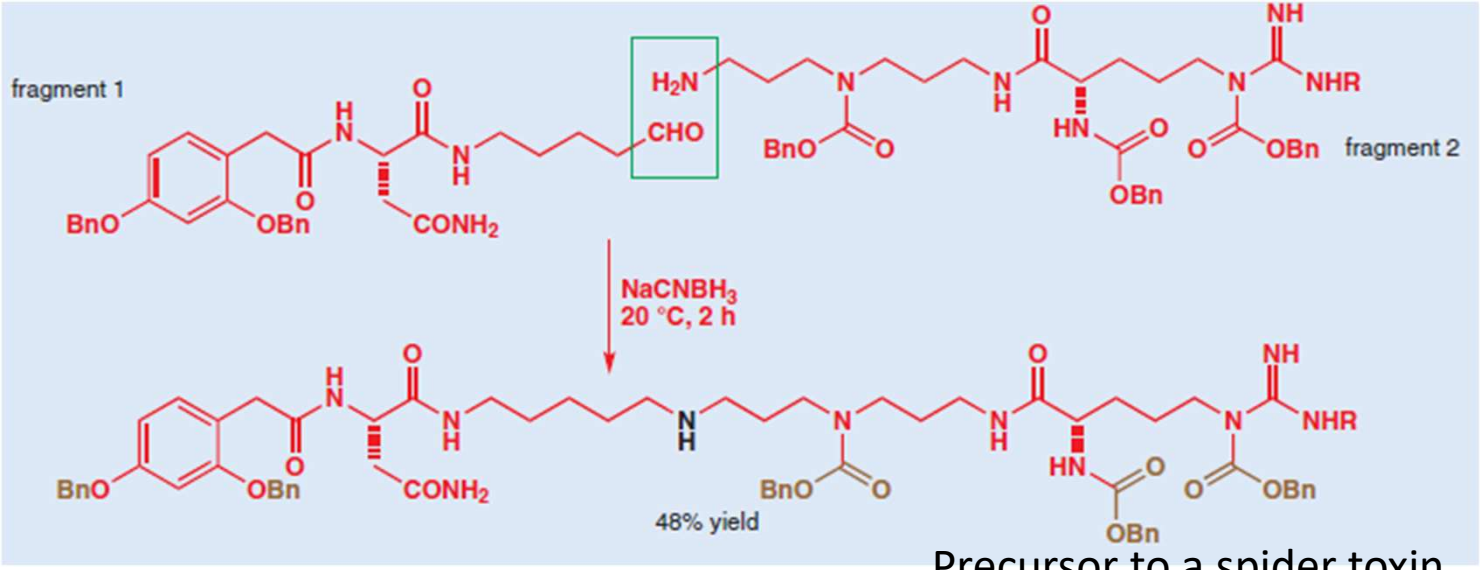


Practice

How would you prepare the following amine?



Practice

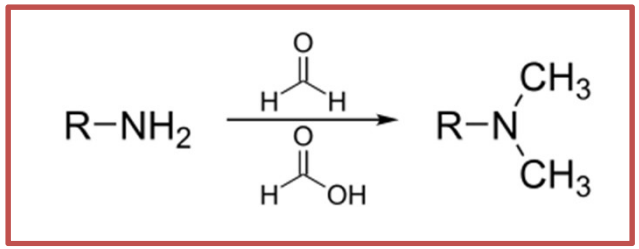


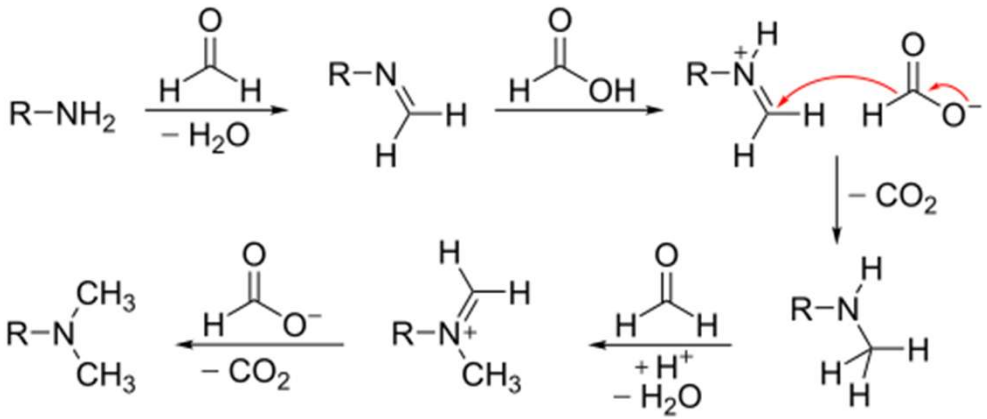
Precursor to a spider toxin

Can you propose a different set of amine and aldehyde?

Formation of methyl amines

Eschweiler–Clarke methylation

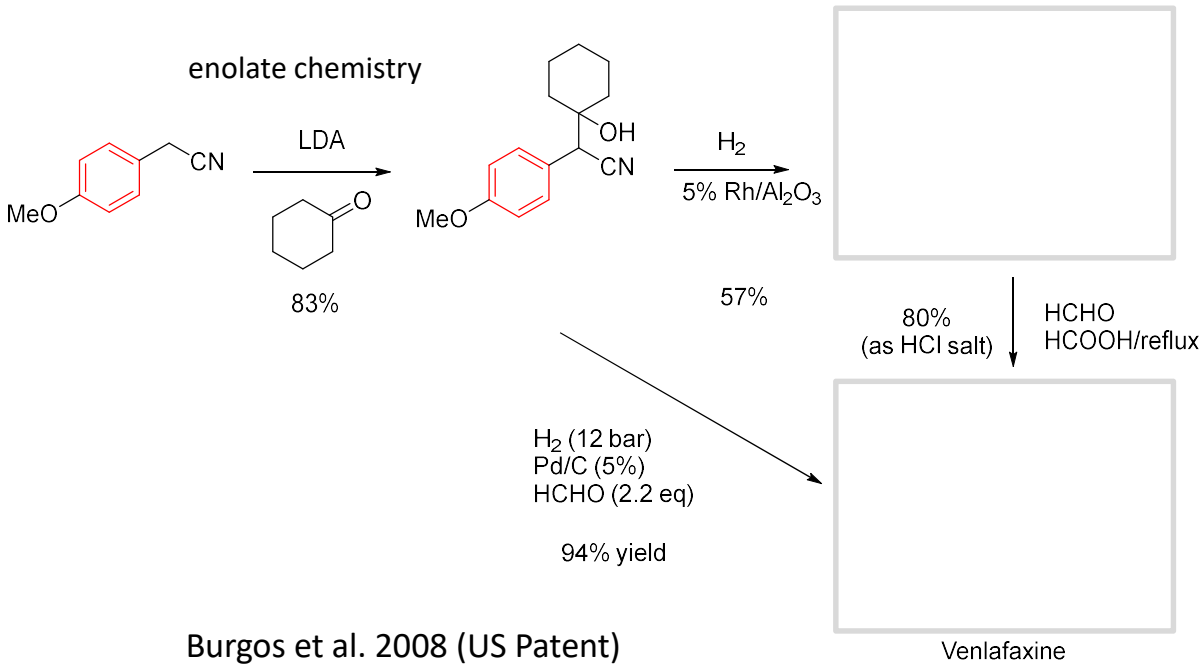




In this reaction, HCOOH is the hydride source.

Synthesis of Venlafaxine

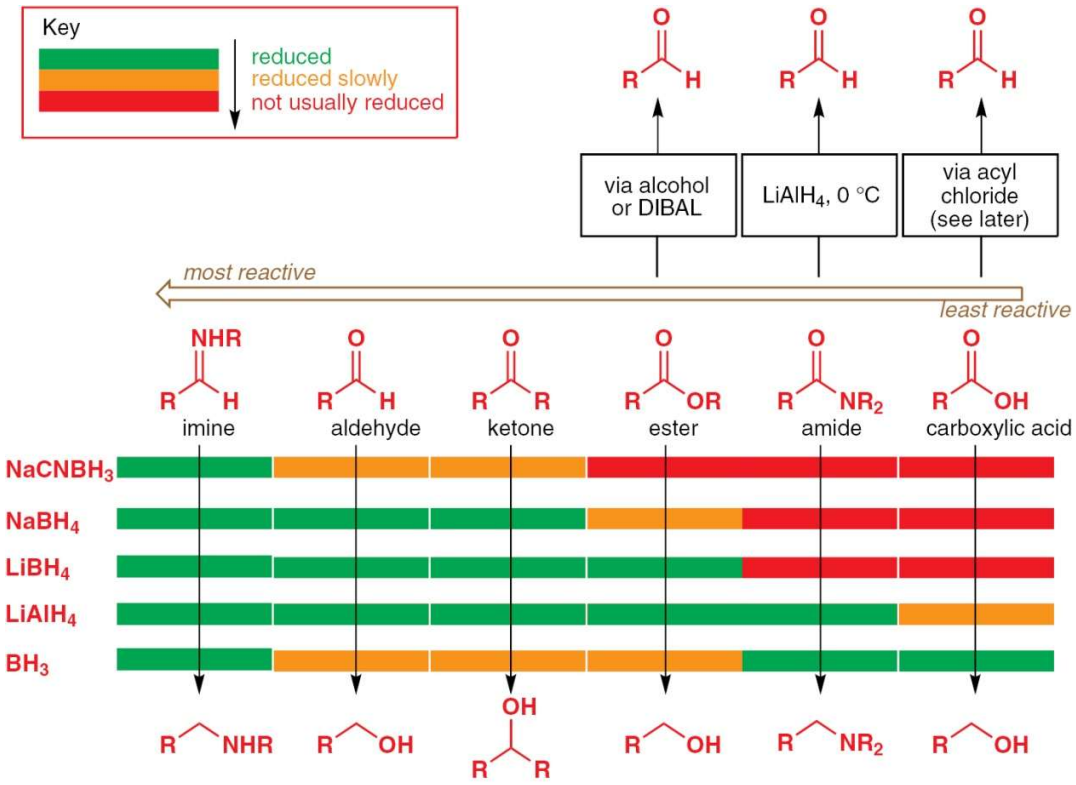
Venlafaxine is an antidepressant drug. It is thought to work by increasing the levels of mood-enhancing chemicals called serotonin and noradrenaline in the brain.



Burgos et al. 2008 (US Patent)

https://www.nhs.uk/medicines/venlafaxine/about-venlafaxine/#:~:text=Venlafaxine%20is%20from%20of%20a,sometimes%20anxiety%20and%20panic%20attacks.

2. Reduction with metal hydrides: overview



Sodium cyanoborohydride ($NaCNBH_3$)

- commonly used for reductive amination
- stable in aqueous solutions even at pH >3

Why not using $NaBH_4$ for reductive amination?

Reduction with metal hydrides: overview

Reducing agents

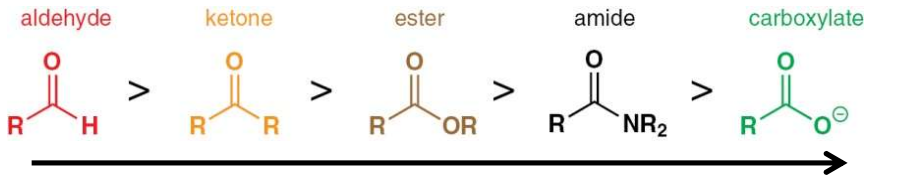
Strong (less selective)

$LiAlH_4$
$LiBH_4$
$NaBH_4$
$NaBH_3CN$

Mild reagent (more selective)

Specific reagents:
- BH_3
- DIBAL

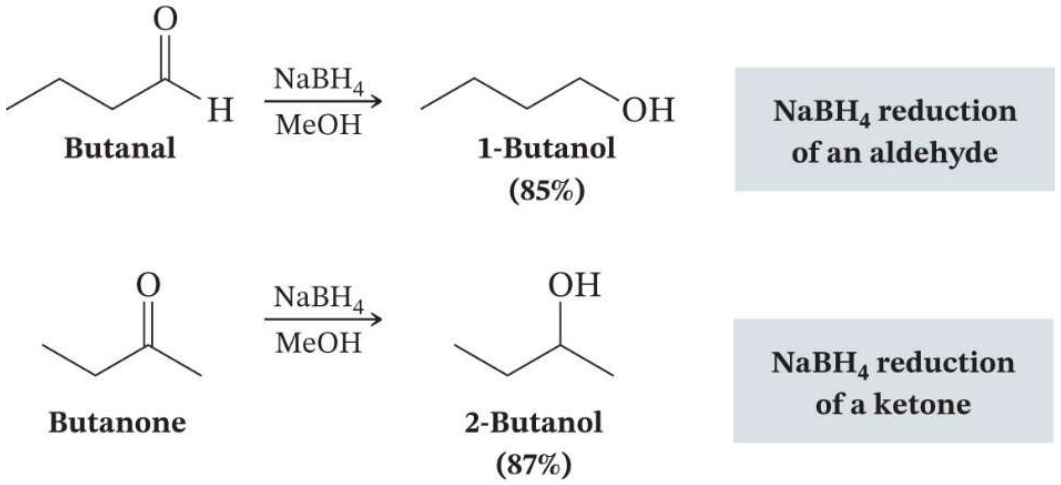
Reactivity of the carbonyl compounds



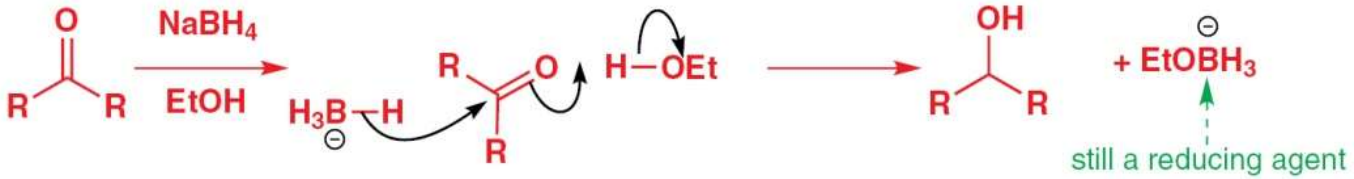
2.1 Common metal hydrides: $NaBH_4$, $LiBH_4$ and $LiAlH_4$

a. $NaBH_4$

- Reaction with **aldehydes/ketones** to produce alcohols.
- Mild and probably most commonly used reducing agent.
- Reactions are performed in ethanol or methanol solvent with an excess of reagents.



- Mechanism: "hydride" addition



In aqueous solvents, sodium borohydride reacts with ionizable alkyl halides (and other derivatives) to give the corresponding hydrocarbon. The reaction mechanism appears to involve the capture of the carbonium ion formed in the solvolysis by the borohydride anion. The reaction has proven very useful for exploring the structure of carbonium ions produced in such solvolyses.[12]

5. Solvent Effects [13]

Could the reducing power of these complex hydrides be influenced by the solvent?

Sodium borohydride has a great advantage over lithium aluminum hydride in that it may be used in a much wider range of solvents. Thus it is highly soluble in water, reacting only slowly with the solvent to evolve hydrogen. Even this slow reaction is markedly decreased by the addition of alkali,[4] and stable solutions of sodium borohydride in concentrated sodium hydroxide are commercially available. Such aqueous solutions readily reduce aldehydes and ketones, even in cases where the solubilities of the compounds in the aqueous system are quite limited (12.3).

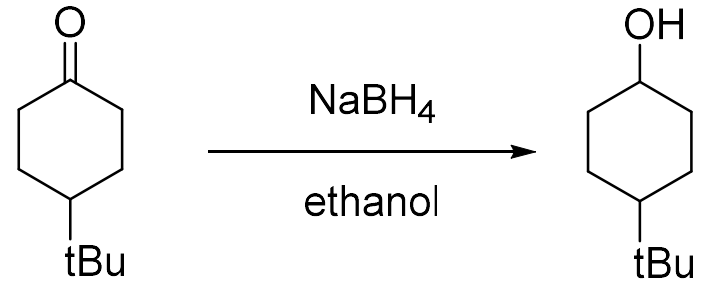
$$4R_2CO + NaBH_4 \longrightarrow Na[B(OCHR_2)_4] \xrightarrow{4H_2O} NaB(OH)_4 + 4R_2CHOH \qquad 12.3$$

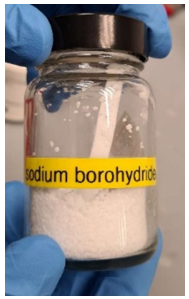
Stoichiometry

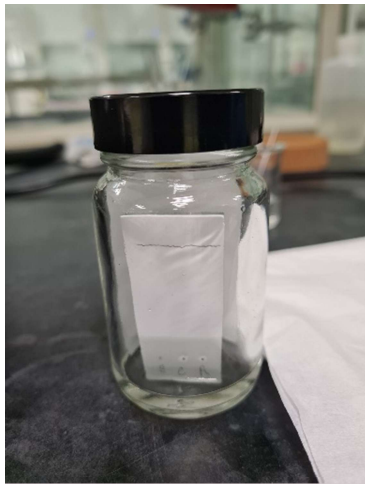
The reducing agent is readily soluble in methyl and ethyl alcohols.[4] It reacts rapidly with methyl alcohol, but only slowly with ethyl alcohol.[13] Thus 100 percent evolution of hydrogen is observed in methanol in 24 minutes at 60°, whereas under the same conditions in ethanol the evolution is less than 2 percent. It is unfortunate that early procedures emphasized the use of methanol as a solvent,[11] so that it is still commonly used, although ethanol possesses the obvious advantage of permitting reductions in homogeneous solution with relatively little loss of the reducing agent through this side reaction with the solvent. **EtOH is a better solvent than MeOH**

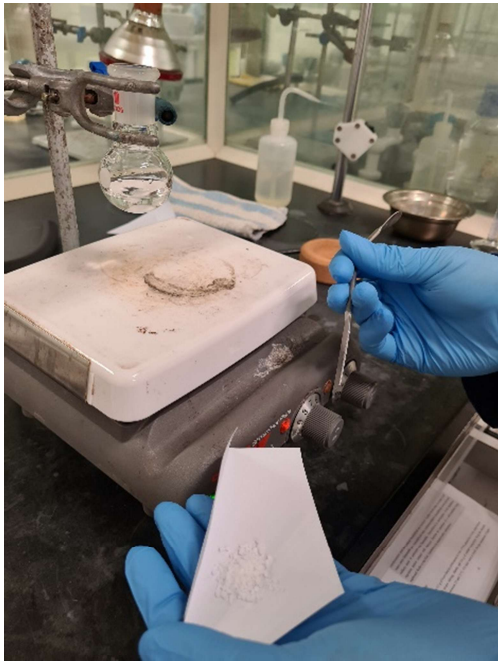
[12] H. M. Bell and H. C. Brown, *J. Amer. Chem. Soc.*, **88**, 1473 (1966).

[13] H. C. Brown, E. J. Mead and B. C. Subba Rao, *J. Amer. Chem. Soc.*, **77**, 6209 (1955).











Sodium borohydride possesses only a modest solubility (0.1 M at 25°) in isopropyl alcohol. However, this solvent possesses the great advantage of being quite stable toward the reagent—standard solutions of sodium borohydride in this solvent have exhibited no measurable change in active hydride over periods of several weeks. Consequently, this solvent not only is convenient for reductions on a preparative scale, but it made possible kinetic studies of the reduction of aldehydes and ketones[14] used in our studies of I-strain (Section VIII-8) and of steric and electronic effects (e.g., Section IX-8).

The great difference in reactivity between benzaldehyde and acetophenone observed in these kinetic studies is noteworthy. Thus, the second-order rate constant for acetophenone at 0° is 2.05×10^{-4} $mole^{-1}$ sec^{-1}, whereas the corresponding constant for benzaldehyde is 820×10^{-4} $mole^{-1}$ sec^{-1}. With a factor of 400 in rate constants, it is evident that aldehyde groups should readily be reduced selectively in the presence of related ketone groupings.

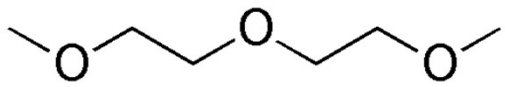
Aldehyde reacts faster than Ketone

so only slightly me (dimethyl ether of diethylene glycol) and triglyme (dimethyl ether of triethylene glycol). The solubility of sodium borohydride in diglyme reaches a maximum (~ 3 M) in the neighborhood of 40°, with greatly reduced solubilities at 0° (0.3 M) and 100° (0.1 M). At 0° a 1:1 solvate with diglyme separates from the solution; at 100° the unsolvated salt separates. Consequently, this medium is a convenient one for the preparation of sodium borohydride of high purity. In triglyme the solubility is high (~ 3 M) and does not vary greatly with temperature.

These solutions in diglyme and triglyme have proven very useful in utilizing sodium borohydride for the generation of diborane[15] and for the hydroboration of olefins.[16,17] However, instead of enhancing the reducing power of sodium borohydride, these solvents appear to decrease it. No significant reduction of acetone was observed in diglyme solution under conditions where the ketone is quantitatively reduced in aqueous or alcohol solution in a matter of minutes.[13]

Aldehydes are reduced by solutions of sodium borohydride in diglyme,[18] so that such solutions may be effective for the selective reduction of aldehydes in the presence of ketones.

Aldehydes can be reduced by sodium borohydride in diglyme, but not ketones.



Diglyme is an aprotic solvent.

activity.

Lithium aluminum hydride is so reactive that possible solvents are essentially restricted to hydrocarbons, ethers, and tertiary amines. It is insoluble in hydrocarbons and readily soluble in ethers. Unfortunately, no detailed study of its reactions in amine solvents is available. It is generally utilized in ethyl ether, tetrahydrofuran, and diglyme, in which it is a powerful reducing agent, with no significant difference apparent in its reducing power.

6. Effect of Different Cations [12, 19]

Early observations indicated a marked difference in the reactivity of lithium and sodium borohydrides. Thus sodium borohydride reduces typical esters (such as ethyl acetate and ethyl benzoate) only very slowly, whereas lithium borohydride reduces such esters quite easily.[20]

In aqueous solution there is no measurable difference in the rate constants for the reactions of sodium and lithium borohydrides with

[14] H. C. Brown, O. H. Wheeler and K. Ichikawa, *Tetrahedron*, **1**, 214 (1957): H. C. Brown and K. Ichikawa, *J. Amer. Chem. Soc.*, **84**, 373 (1962).

[15] H. C. Brown and P. A. Tierney, *J. Amer. Chem. Soc.*, **80**, 1552 (1958).

[16] H. C. Brown and B. C. Subba Rao, *J. Amer. Chem. Soc.*, **78**, 2582 (1956); **81**, 6423 (1959).

[17] H. C. Brown and B. C. Subba Rao, *J. Org. Chem.*, **22**, 1136 (1957); *J. Amer. Chem. Soc.*, **81**, 6428 (1959).

[18] H. C. Brown and G. Zweifel, unpublished observations.

[19] H. C. Brown and B. C. Subba Rao, *J. Amer. Chem. Soc.*, **78**, 2582 (1956).

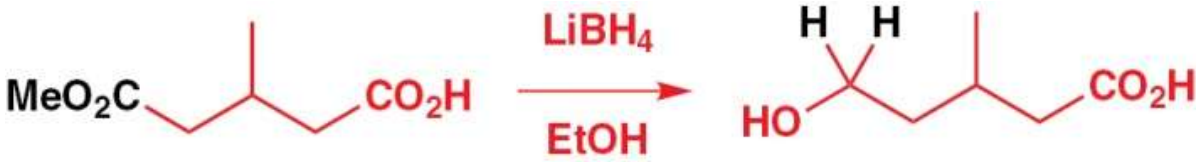
[20] R. F. Nystrom, S. W. Chaikin and W. G. Brown, *J. Amer. Chem. Soc.*, **71**, 3245 (1949).

Common metal hydrides: $NaBH_4$, $LiBH_4$ and $LiAlH_4$

b. $LiBH_4$:

- $LiBH_4$ is commonly used for the selective reductions of **esters** and lactones to the corresponding alcohols in the presence of carboxylic acids, amides and nitriles.
- **Aldehydes, ketones** can be reduced.

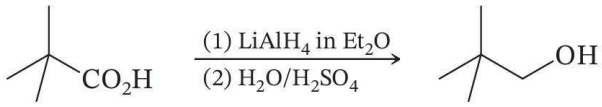
- Stronger than $NaBH_4$
- Milder than $LiAlH_4$



Common metal hydrides: $NaBH_4$, $LiBH_4$ and $LiAlH_4$

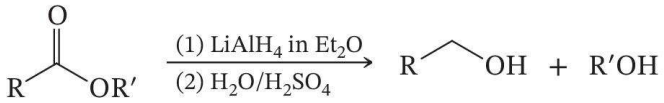
c. $LiAlH_4$: Strong reducing agent

- Reduce **carboxylic acids, esters, aldehydes** to primary alcohols, ketones to secondary alcohols



2,2-Dimethylpropanoic acid → **Neopentyl alcohol (92%)**

$LiAlH_4$ reduction of a carboxylic acid



$LiAlH_4$ reduction of an ester



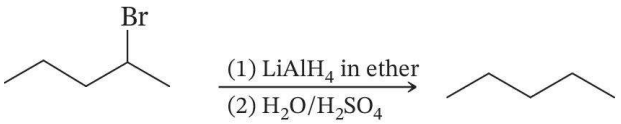
Reproduced with permission from Millipore-Sigma

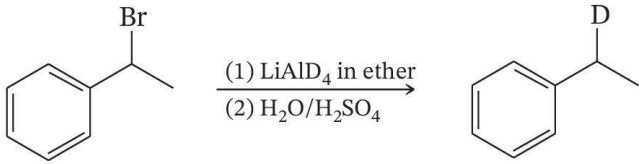


Pyrophoric chemical

$LiAlH_4$

Can use to remove halogen (reduction of alkyl halides).

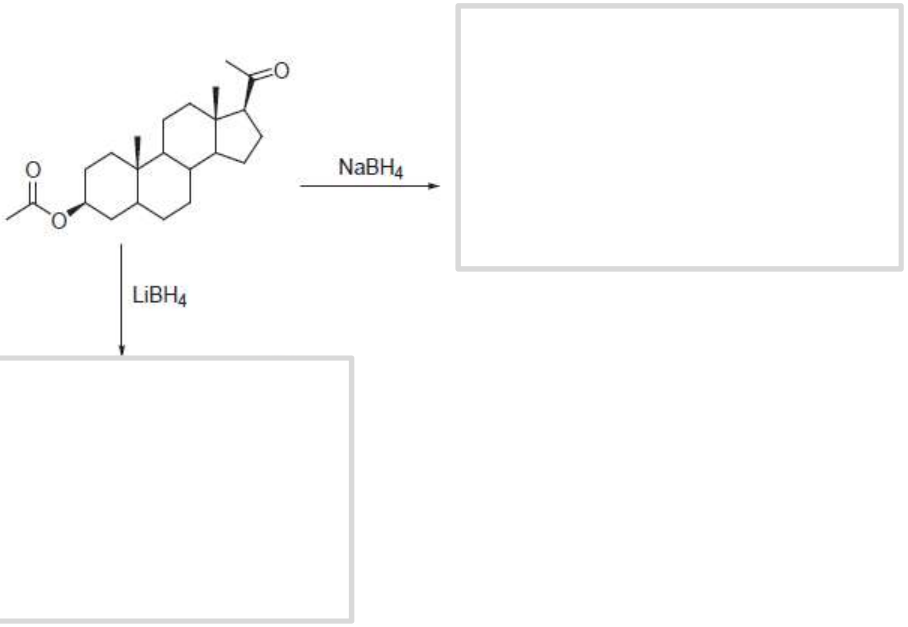




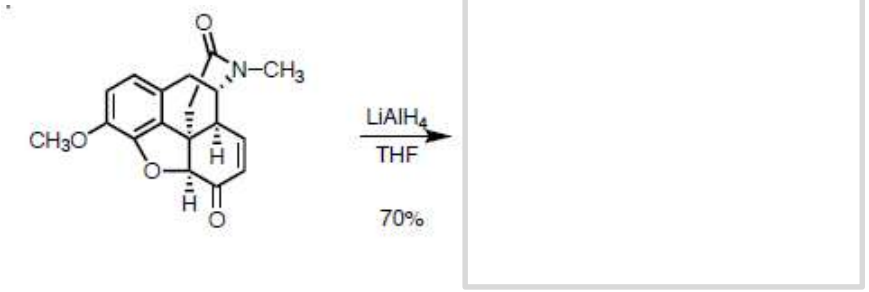
Alternative approaches for reduction of alkyl halides:

a. Elimination reaction followed by hydrogenation
b. Through formation of alkyl radicals

Practice questions



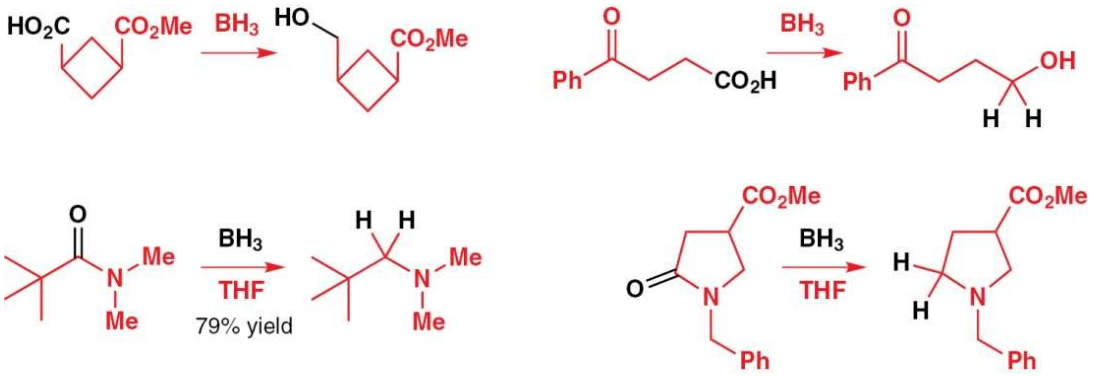
Practice questions



Borane

Borane BH_3 (borane is a gas, the actual reagents are $BH_3.Et_2O$ or $BH_3.Me_2S$)
Borane accept electron pair into the boron's empty p orbital, thus it react selectively with electron rich carbonyl groups: **carboxylic acids** to form primary alcohols and **amides** to form amines.

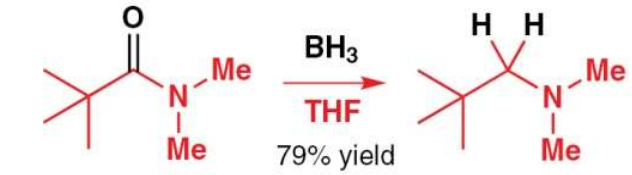
Borane reduces esters and ketones much slower than carboxylic acids. Reactions can be selective.

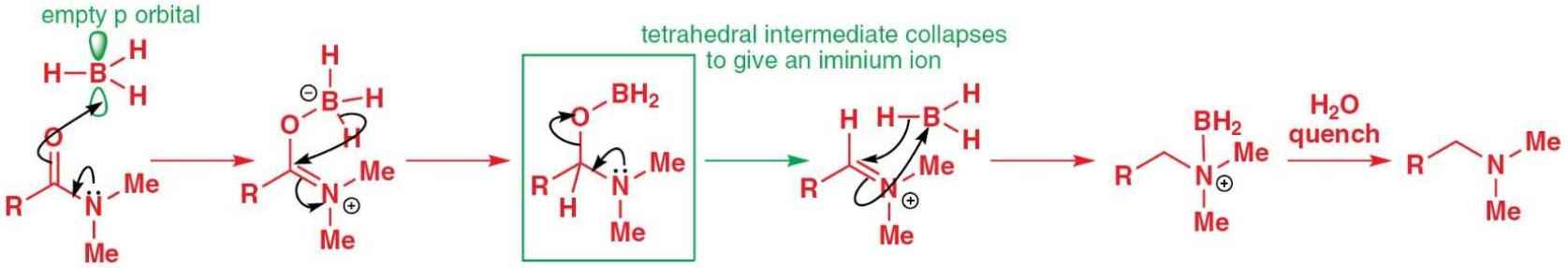


Comparing with $LiBH_4$



BH_3 reduction mechanism



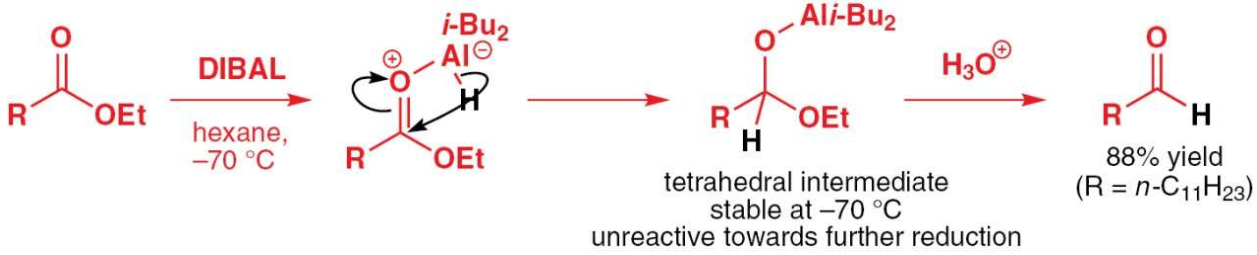


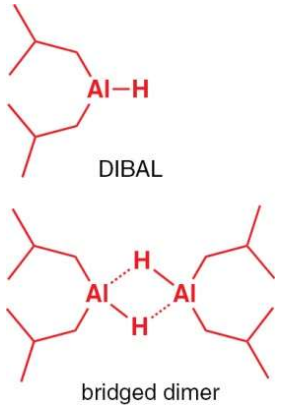
Esters and ketones are less Lewis basic than amides and carboxylates.

DIBAL-H

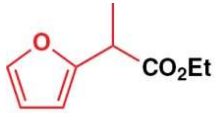
DIBAL-H (diisobutyl aluminium hydride, which is an alane):

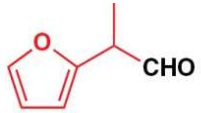
Partial Reduction (**esters to aldehyde**) at low temperature





Question: what is the other way to convert esters to aldehydes?

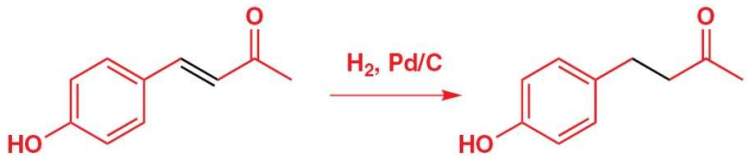

readily available


required for synthesis of
antibiotic monensin

3. Catalytic hydrogenation (H_2/Metal)

3.1 Hydrogenation of alkenes

Alkenes react with H_2 in the presence of Pd (usually Pd/C) to give alkanes



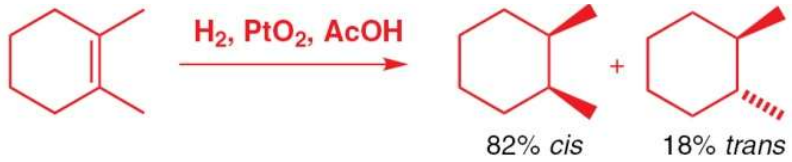
Hydrogen (H_2) non-polar: in combination with a metal catalyst, hydrogen can reduce C=C, C=N, triple bonds etc.

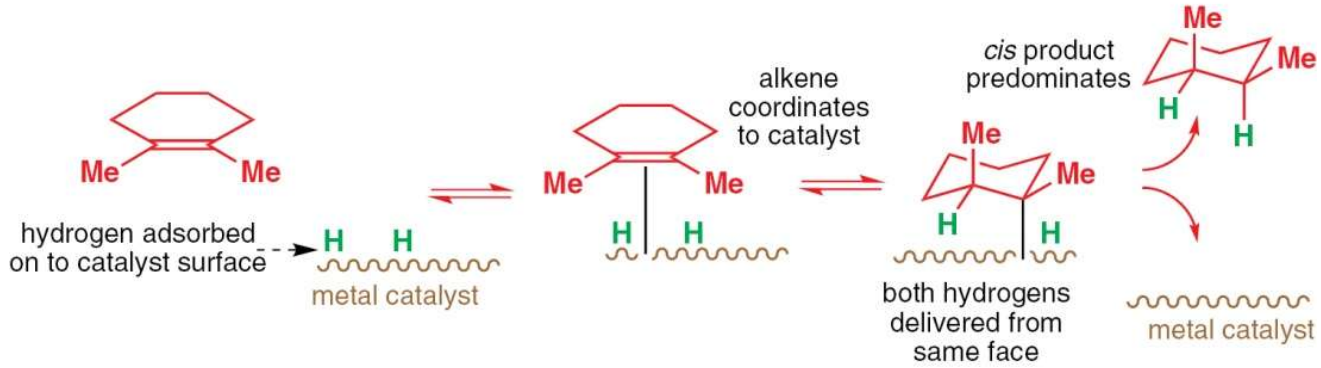
Catalysts:
- Palladium on charcoal (Pd/C, usually 5% or 10% by mass). It can be recycled because the Pd/C is insoluble and can be recovered by filtration.
- PtO_2 (Adams' catalyst) – The actual catalyst is not the platinium oxide, but Pt metal during hydrogenation.



Hydrogenation of alkenes

Mechanism: take place on the metal surface, syn-addition

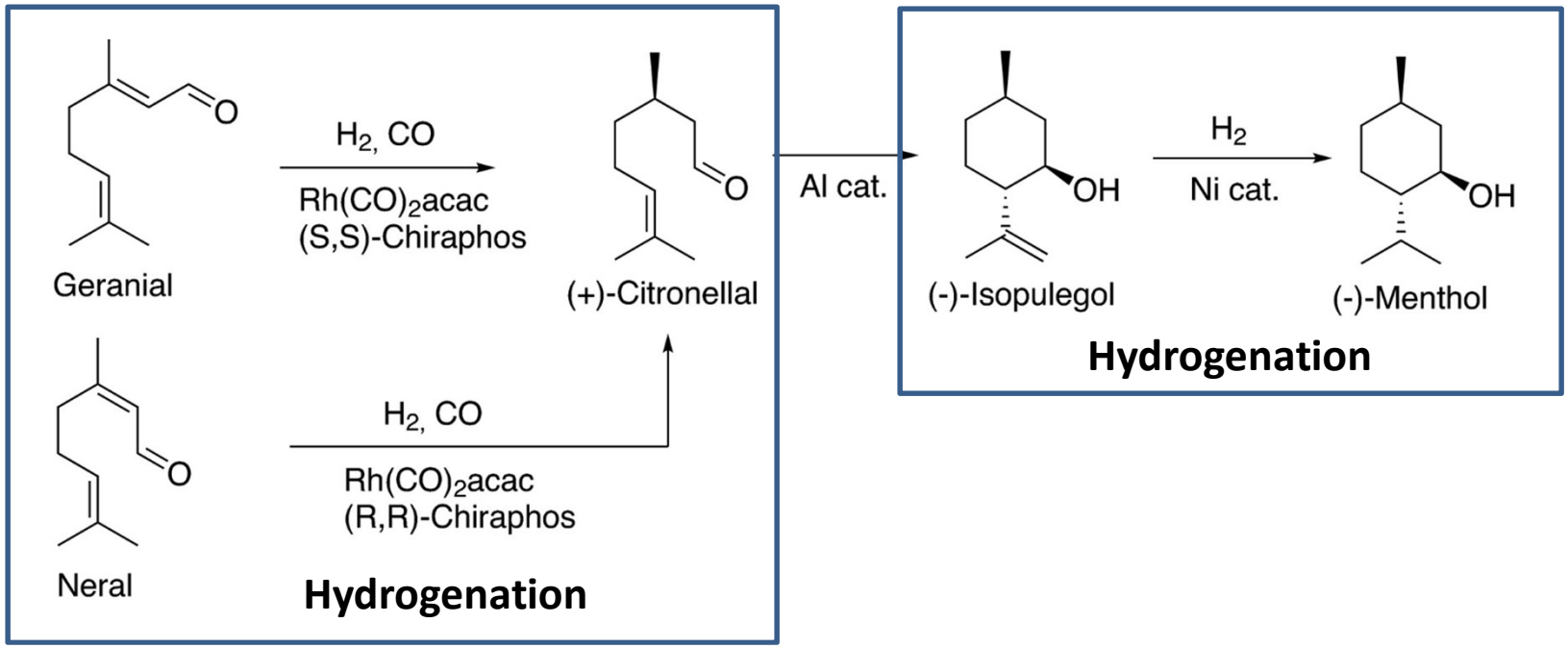




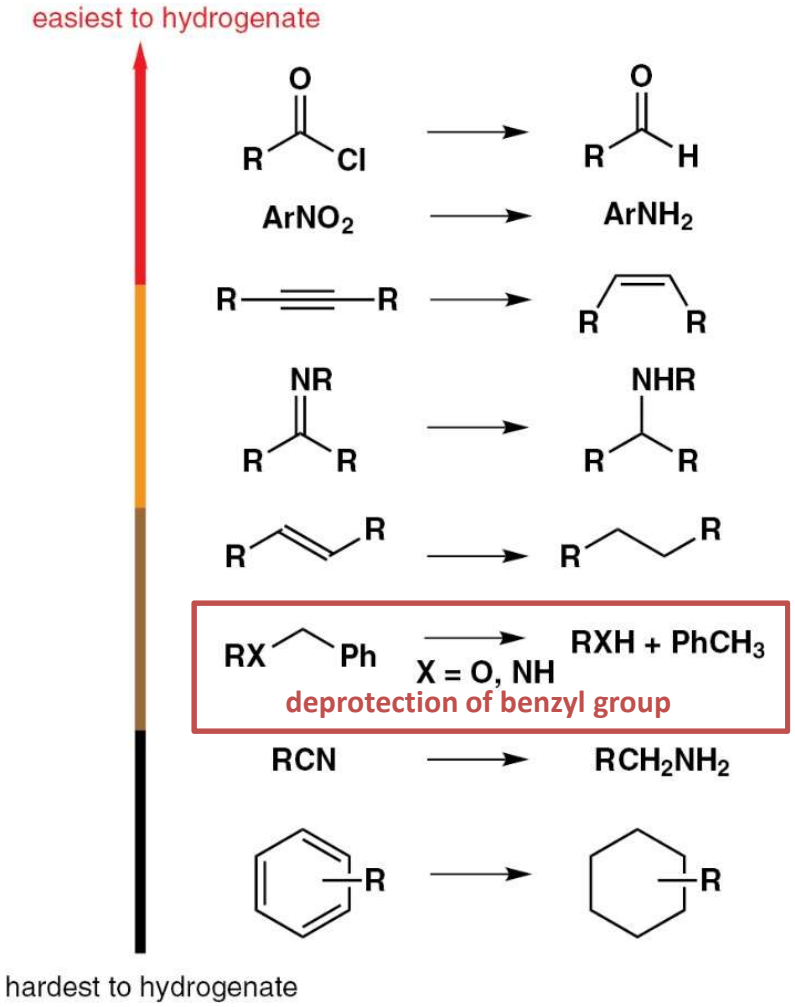
Aromatic ring can be reduced



BASF (–)-menthol process



3.2 Hydrogenation of other functional groups

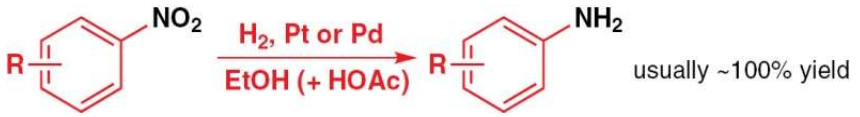


Acid chlorides

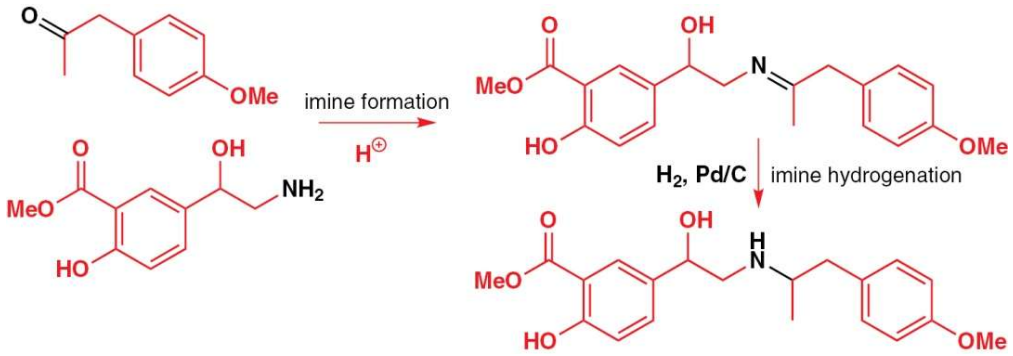


(carboxylic acid→ acid chloride→ aldehyde)

Nitro groups



Imines (reductive amination)



Summary: reduction reactions

1. Preparation of amines
2. Reductive amination
3. Reduction reactions with metal hydrides
   - common reagents
   - specific reagents (Borane, DIBAL-H)
4. Catalytic hydrogenation